

CATEGORY		MID-RANGE AMD MOTHERBOARDS				
FEATURES (30%)	Brand	ASUS	MSI	ASUS	AOpen	Gigabyte
	Model	A7N8X	K7N2 Delta FISR	A7V600	AK77-8X	GA-7NNXP
	Chipset	Nforce2 400-Ultra	Nforce2 400-ultra	VIA KT600	VIA KT400	Nforce2 400-Ultra
	Interface type	Socket A	Socket A	Socket A	Socket A	Socket A
PERFORMANCE (70%)	Maximum CPU supported	Athlon 3200+	Athlon 3200+	Athlon 3200+	Athlon 2800+	Athlon 3200+
	Memory speed (MHz) / Max memory (GB) / Memory slots	400 / 3 / 3	400 /3/ 3	400 /NA/ 3	333/ 4 /4	400 / 3/ 4
	Form factor	ATX	ATX	ATX	ATX	ATX
	AGP port / Onboard AGP / Type	✓ / ✗ / 8X	✓ / ✗ / 8X	✓ / ✗ / 8X	✓ / ✓ / 8X	✓ / ✗ / 8X-Pro
	PCI slots	5	5	6	5	5
	Onboard Sound / Built in APU	✓ / ✓	✓ / ✓	✓ / ✗	✓ / ✗	✓ / ✓
	Onboard FireWire / Firewire-B	✓ / ✗	✓ / ✗	✗	✗	✓ / ✗
	USB ports / Rear / Ports via cable	6/4/2	6/4/2	8/4/4	8/4/4	8/4/4
	Ethernet / Gigabit Ethernet	✓ / ✓	✓ / ✗	✗ / ✓	✗ / ✓	✓ / ✓
	RAID 0, 1, 0+1	RAID 0, 1	RAID 0, 1	✗	✗	RAID 0, 1, 0+1
	SATA / SATA RAID	✓ / ✓	✓ / ✓	✓ / ✗	✗ / ✗	✓ / ✓
	Overclocking and BIOS updating features / software	✓	✓	✓	✗	✓
	Core voltage adjustment	✓	✓	✓	✓	✓
	Freq adjustment	✓	✓	✓	✓	✓
	Component layout on the motherboard (S05)*	3.17	3.67	2.50	3.00	3.00
	Package contents					
	Manual / CDs	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓	✓ / ✓
	Necessary cables provided	✓	✓	✓	✓	✓
	Gaming Benchmarks: Serious Sam Second Encounter					
	Average fps	98.87	96.73	84.37	69.40	95.67
	UT2003					
	Average fps	111.78	111.23	110.79	117.16	111.64
	Quake III Arena					
	Average fps	203.57	191.67	155.33	135.67	190.00
	Real world Benchmarks					
	ZDbench Content Creation 2004	24.3	25.6	24.3	14.1	26.3
	ZDbench Business Winstone 2004	26.1	24	22.7	20.3	24.3
	Video encoding (Time in secs)	73	75	75	93	72
	Synthetic Benchmarks					
	3D Mark 2003 (patch 340)	3,362	3,341	3,311	3,022	3,367
	3D Mark 2001 SE (patched- for onboard without AGP slot)	11,855	11,642	11,470	10,272	11,659
	PC Mark 2004					
	Memory Index	2,751	2,624	3,704	3,391	2,793
	Disk Index	2,911	2,901	2,390	2,049	2,881
	CPU Index	3,768	3,745	2,843	2,913	3,757
	Graphics	2,860	2,899	3,314	3,350	2,984
	Total Score	3,959	3,926	3,822	3,437	3,957
	Sisoft Sandra 2004 Professional: CPU sub-system					
	CPU Dhrystone / Whetstone	8,341 / 3,437	8,297 / 3,428	8,349 / 3,444	7,894 / 3,256	8,293 / 3,446
	Multimedia - CPU Integer / FPU SSE	20,027 / 20,571	19,965 / 20,515	20,010 / 20,529	18,915 / 19,419	20,062 / 20,606
	Memory sub-system					
	ALU / RAM	3,043	2,818	2,701	2,121	3,047
	FPU / RAM	2,860	2,645	2,516	2,006	2,864
	Disk drive					
	Drive Index MBps	26,304	26,254	29,204	31,742	26,274
OVERALL SCORE	Performance	58.50	57.35	54.89	46.52	58.24
	Features	20.27	19.47	17.50	15.20	21.70
	Grand Total (Performance + Features)	78.77	76.82	72.39	61.72	79.94
	VFM	9.00	8.54	11.87	10.83	6.15
	Price	8,750	9,000	6,100	5,700	13,000
	Grades					

* Scale of 5

allow gameplay on par with the nForce2-based boards, especially at higher resolutions.

In the real-world test, the competition was cut-throat, with the Gigabyte 7NNXP beating ASUS by a hair's breadth. MSI's K7N2 Delta was also in the league, falling short by a few paces, and coming in at third spot. The ASUS A7V600, too, put up a commendable performance. The AOpen board, straddled with the KT400 chipset, was badly hit in the real-world test, and its scores were low. This can be attributed to the underpowered

memory subsystem of the KT400 chipset, which runs in single-channel. The nForce2 has dual-channel memory architecture, whereas the new VIA KT600 uses a different approach towards bettering the memory implementation. The results are visible—the KT600-based board performed about as well as the nForce2-based boards in the real-world test.

In the synthetic tests, especially the I/O test, the VIA Southbridge extracts the maximum out of your hard disk, and is therefore a clear winner in terms of I/O. So, if your work involves